CSET485 — ASSIGNMENT 1

STATISTICAL ANALYSIS, MODEL EVALUATION AND ALGORITHMIC FAIRNESS

Dataset : COMPAS Recidivism Dataset

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Course Code : CSET485

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1. Introduction

In this assignment, I worked with the COMPAS recidivism dataset to understand how statistical models make predictions and how fair those predictions really are. The main goal was to see how different models behave, how the results change when we look at different demographic groups, and how the assumptions we make can affect the conclusions we draw.

First, I compared the classifications from linear regression and logistic regression. Then I ran the regression separately for different subgroups to see whether the results stayed the same. I also looked at omitted-variable bias, which shows how leaving out an important variable can change the results. After that, I tested a selected set of 50 adversarial cases to see where the model struggles. In the last part, I checked fairness using three measures: demographic parity, equalized odds, and predictive parity, at different decision thresholds.

One thing I learned is that accuracy alone is not enough to judge a model. Things like how the data is preprocessed, which threshold is chosen, how the model performs for each subgroup, and what happens when it predicts wrongly all matter a lot. Since this is observational data, I have also tried to keep in mind the uncertainty in the results and the possibility of confounding.

Overall, this assignment helped me understand the strengths and limitations of these models, and it made me think about how much evidence we would need before using recidivism predictions to make serious real-world decisions.

2. Dataset and Methodology

2.1 Dataset Overview

For this assignment, I used the COMPAS recidivism dataset to study how statistical models predict whether a person will reoffend within two years. The dataset has demographic details, criminal history variables, and the final recidivism outcome. Using these, I could check how things like prior records, age, and demographic groups are related to the predicted outcome.

2.2 Model Comparison

In the first part, I compared Linear Regression and Logistic Regression. I took 2,165 test cases and classified them using the thresholds given in the notebook. Then I checked how often the two models agreed and where they gave different results.

2.3 Subgroup Regression

Here I fitted logistic regression on the full dataset and then separately on the African-American and Caucasian subgroups. I used prior criminal-record count and age as the predictors for two-year recidivism. To understand the results, I looked at the coefficients, confidence intervals, and p-values.

2.4 Omitted-Variable Bias

In this experiment, I removed some predictors from the regression models to see how the estimated relationships changed. The variables I worked with were age, juvenile felony count, juvenile misdemeanor count, and juvenile other-offence count.

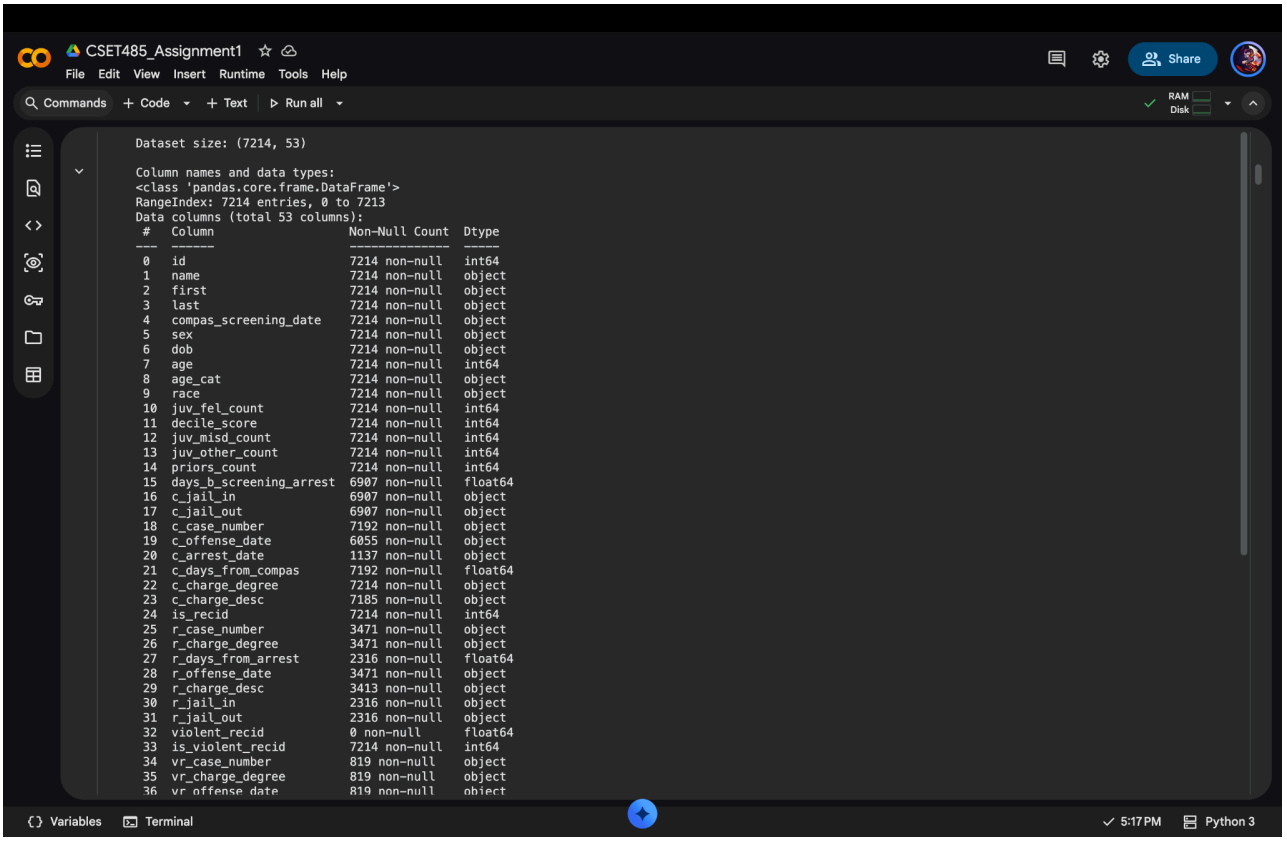
2.5 Adversarial Subset Analysis

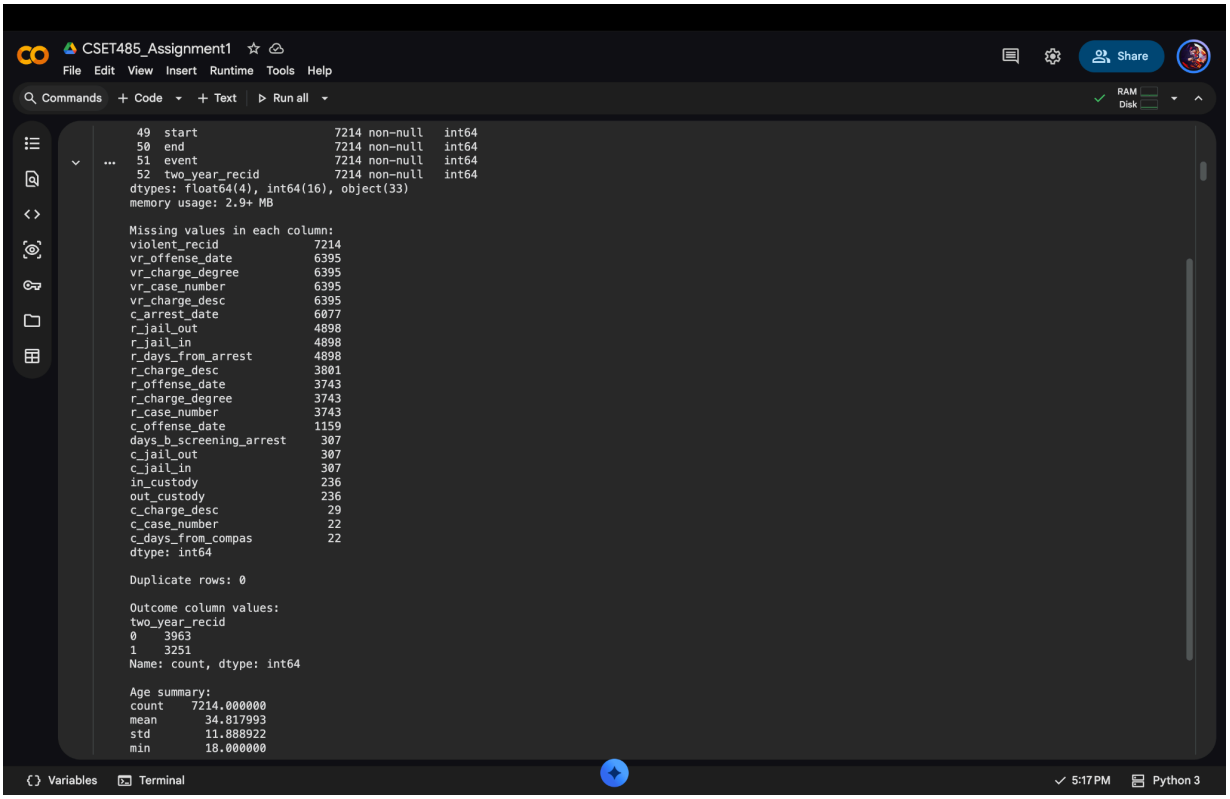
For this part, I picked a subset of 50 cases and studied its characteristics, mainly the prior criminal-record counts. Since the sample is small and was specially selected, I was careful while interpreting it because it may not represent the full dataset.

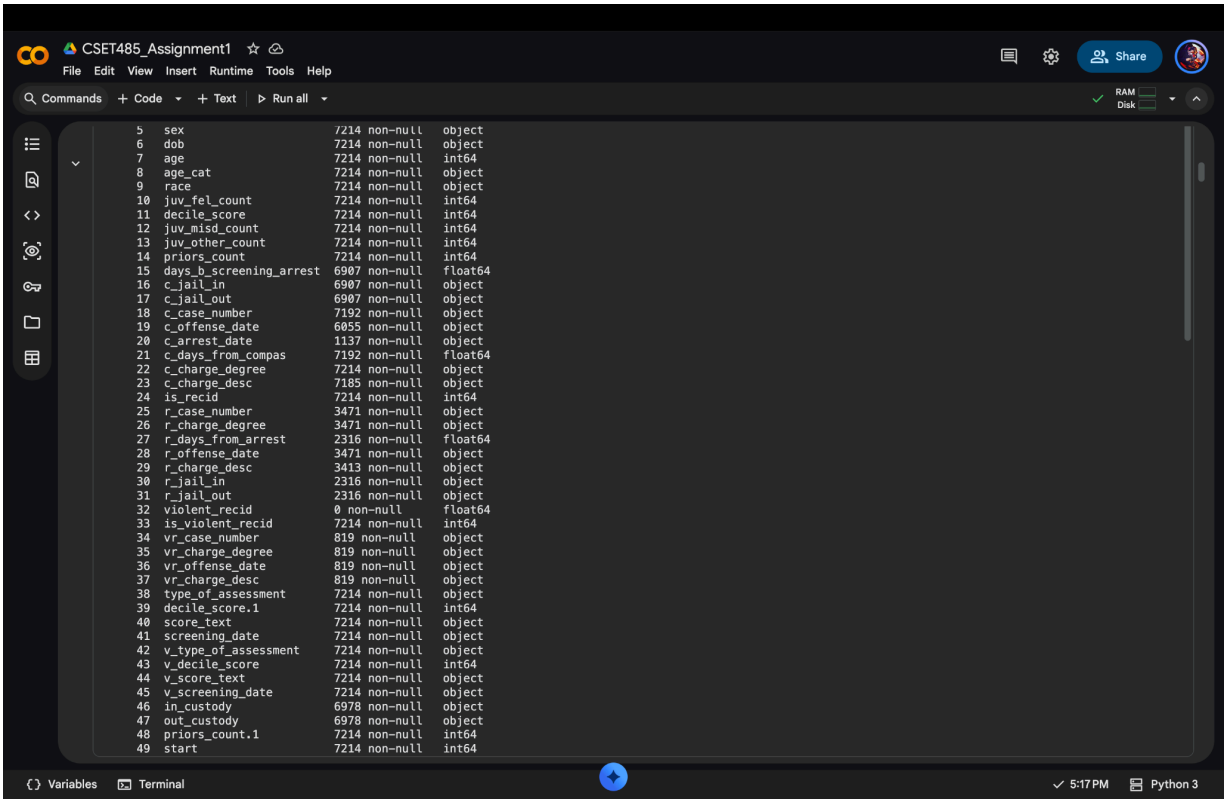
2.6 Fairness Evaluation

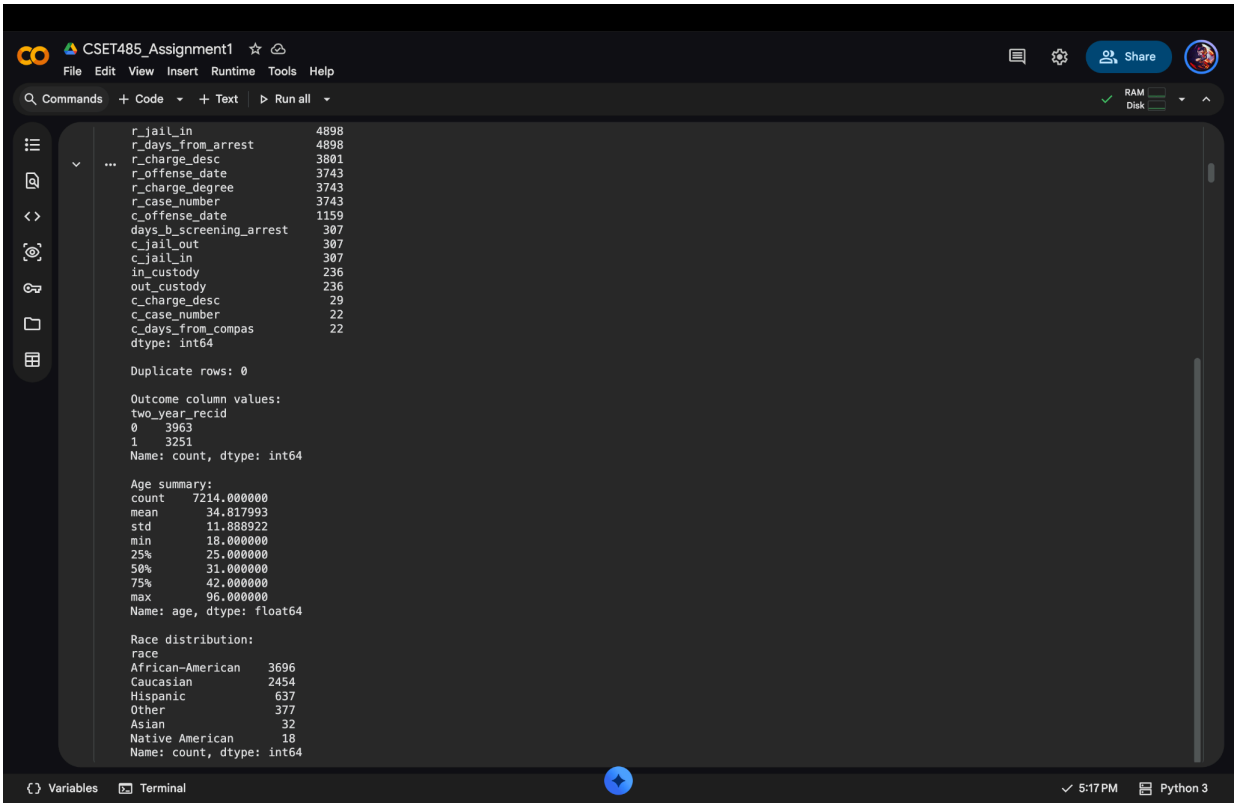
In the last analysis, I compared demographic parity, equalized odds, and predictive parity at decision thresholds from 0.05 to 0.95. The aim was to see how changing the threshold affects the fairness measures and what trade-offs come up when a model is actually used.

Screenshots :-




CSET485_Assignment1
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
RAM Disk
49 start 7214 non-null int64
50 end 7214 non-null int64
51 event 7214 non-null int64
52 two_year_recid 7214 non-null int64
dtypes: float64(4), int64(16), object(33)
memory usage: 2.9+ MB
Missing values in each column:
violent_recid 7214
vr_offense_date 6395
vr_charge_degree 6395
vr_case_number 6395
vr_charge_desc 6395
c_arrest_date 6077
r_jail_out 4898
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r_days_from_arrest 4898
r_charge_desc 3801
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c_offense_date 1159
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c_jail_in 307
in_custody 236
out_custody 236
c_charge_desc 29
c_case_number 22
c_days_from_compas 22
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Outcome column values:
two_year_recid
0 3963
1 3251
Name: count, dtype: int64
Age summary:
count 7214.000000
mean 34.817993
std 11.888922
min 18.000000
Variables Terminal
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CSET485_Assignment1
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Commands + Code + Text Run all
RAM Disk
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20 c_arrest_date 1137 non-null object
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22 c_charge_degree 7214 non-null object
23 c_charge_desc 7185 non-null object
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33 is_violent_recid 7214 non-null int64
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3. Results and Discussion

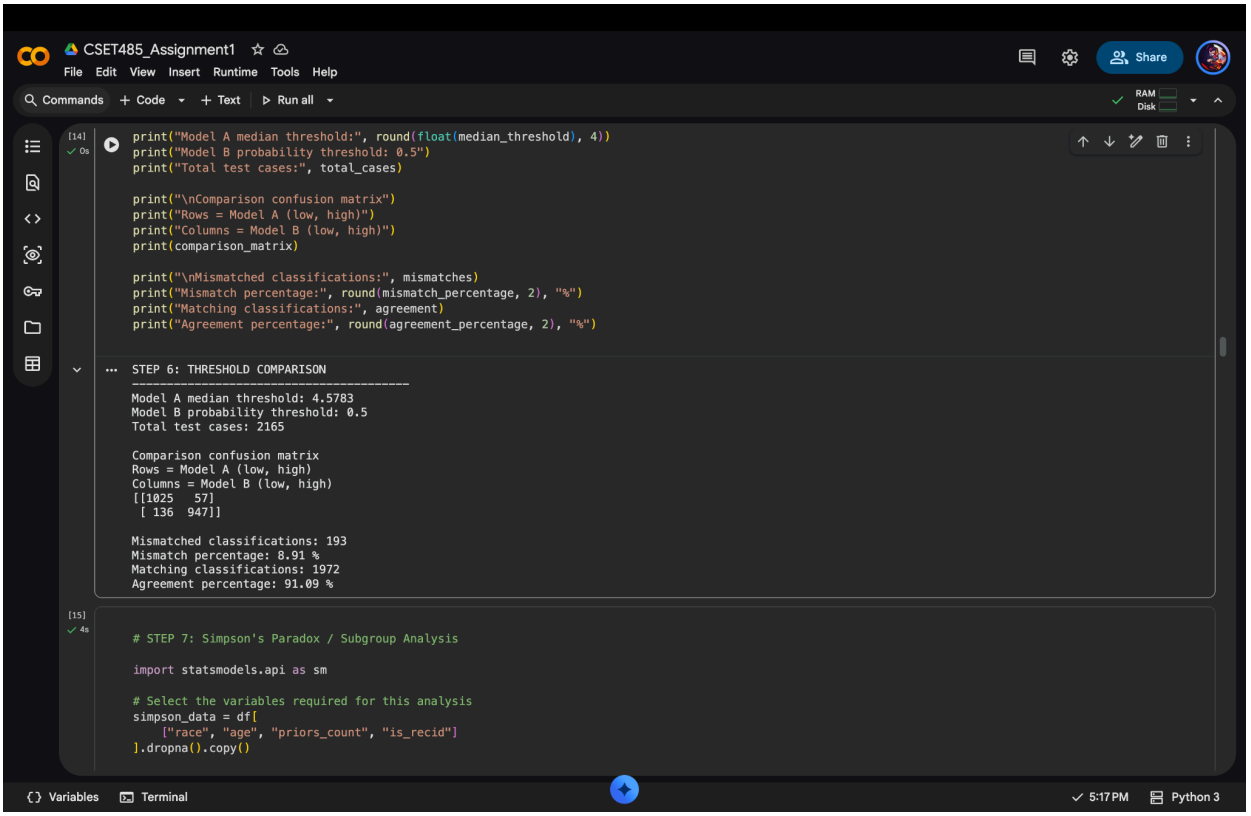
3.1 Comparison of the Two Models

I compared Linear Regression and Logistic Regression on 2,165 test cases. Both models gave the same classification for 1,972 cases, which is 91.09% agreement. The remaining 193 cases (8.91%) were classified differently.

From the comparison matrix, 57 cases were low according to Model A but high according to Model B. The other 136 cases were high according to Model A but low according to Model B. This shows that the choice of model and threshold can change the prediction for an individual person.

Linear Regression predicts a numerical value, while Logistic Regression gives a probability for a yes/no outcome. So their scales and thresholds don't mean the same thing. I think these mismatched cases should be studied properly before anyone uses such predictions for important decisions.

Screenshot :-



Model Comparison Output and Confusion Matrix

The output compares the classifications produced by Linear Regression and Logistic Regression on 2,165 test cases. The models agreed on 1,972 cases (91.09%) and disagreed on 193 cases (8.91%). The comparison confusion matrix shows the distribution of matching and mismatching classifications.

3.2 Subgroup Regression Results

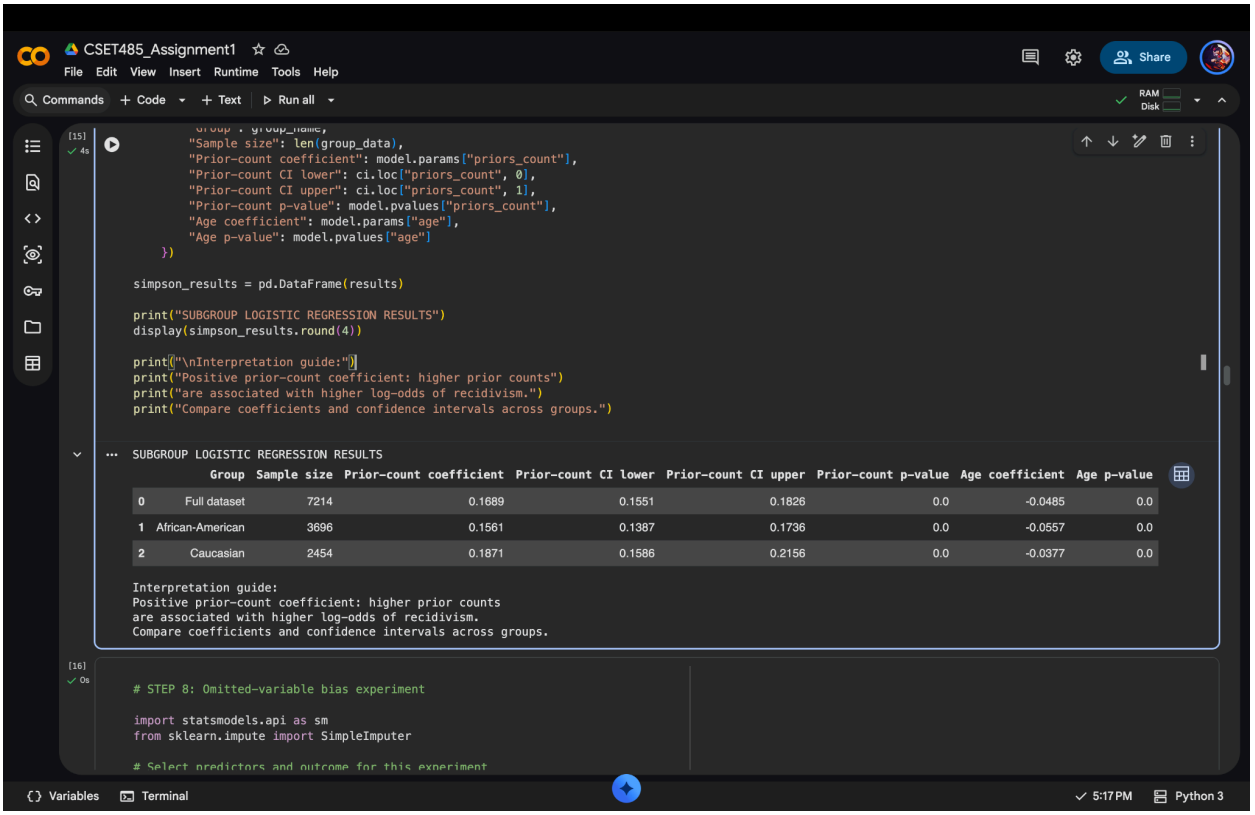
The full dataset had 7,214 observations. The African-American subgroup had 3,696 observations and the Caucasian subgroup had 2,454.

The prior-count coefficient was 0.1689 for the full dataset (95% CI: [0.1551, 0.1826]). For the African-American subgroup it was 0.1561 (CI: [0.1387, 0.1736]), and for the Caucasian subgroup it was 0.1871 (CI: [0.1586, 0.2156]).

All three coefficients are positive and none of the confidence intervals include zero. This means that in these models, a higher number of prior records is associated with higher

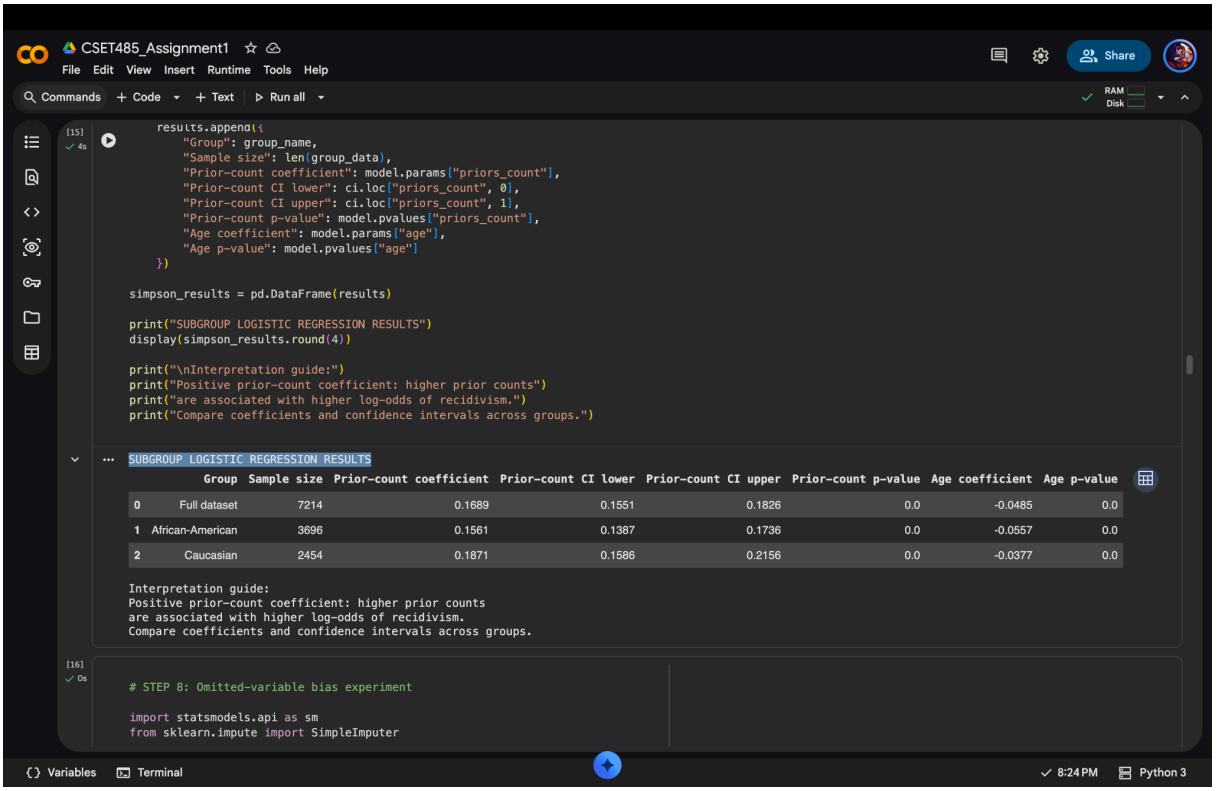
log-odds of recidivism. But this only shows association, not causation, and it does not prove that the difference between the two subgroups is statistically significant.

Screenshot :-



Subgroup Logistic Regression Results

The table compares logistic regression results for the full dataset, African-American subgroup, and Caucasian subgroup. The prior-count coefficients were 0.1689, 0.1561, and 0.1871, respectively. All three coefficients were positive, and their reported 95% confidence intervals excluded zero, indicating positive associations between prior-count values and the log-odds of recidivism in these fitted models. These results do not establish causation.



Subgroup Logistic Regression Results with 95% Confidence Intervals

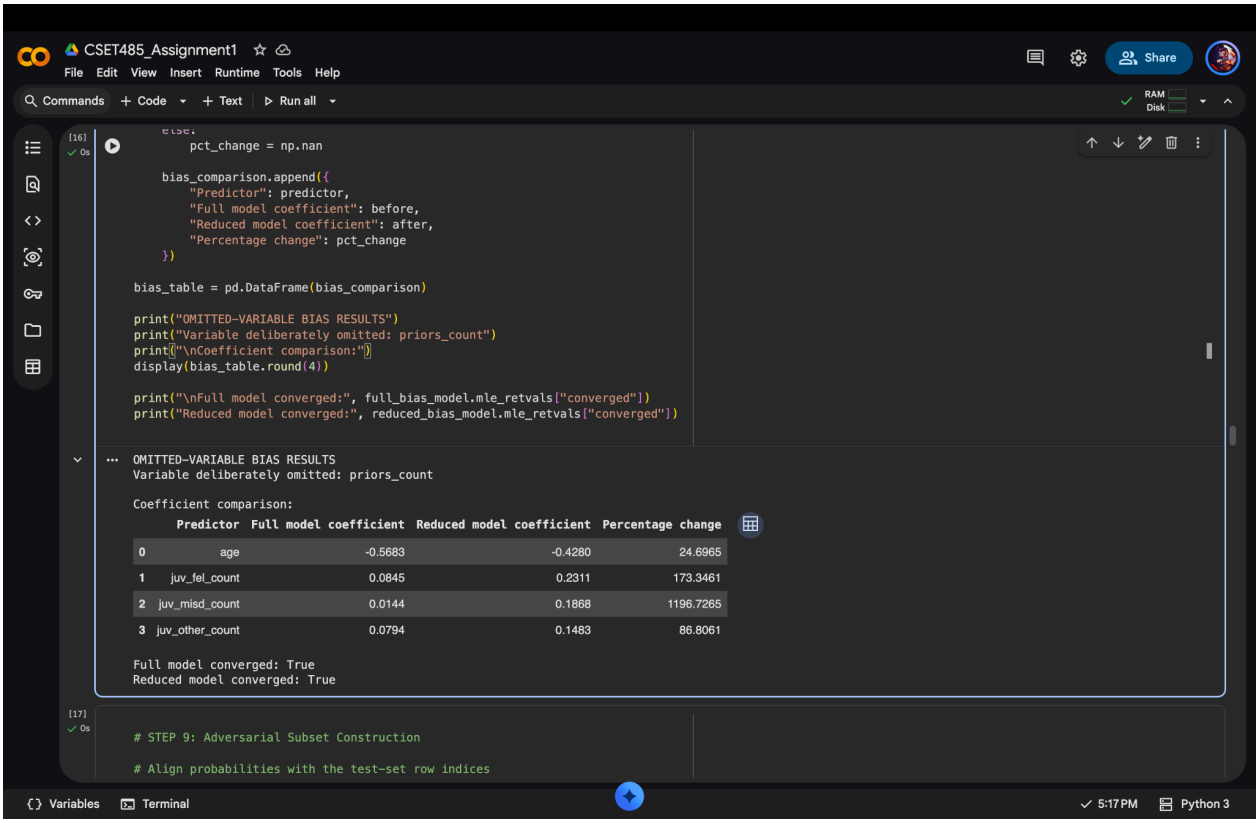
The table presents logistic regression results for the full dataset and the African-American and Caucasian subgroups. The prior-count coefficients are positive, and the reported 95% confidence intervals exclude zero. This indicates a positive association between prior-record count and the log-odds of recidivism in each fitted model. The results show association, not causation.

3.3 Omitted-Variable Bias

In this experiment, I checked how the regression estimates changed when some predictors were left out. The four predictors were age, juvenile felony count, juvenile misdemeanor count, and juvenile other-offence count.

The biggest change was 1196.7265%, for juvenile misdemeanor count. This means its coefficient was very sensitive to which variables were in the model. However, a big percentage change does not automatically prove bias, especially when the original coefficient is very close to zero. To judge it properly, we should also look at the sign, confidence interval, and significance of the coefficients.

Screenshot :-



Omitted-Variable Bias Experiment Results

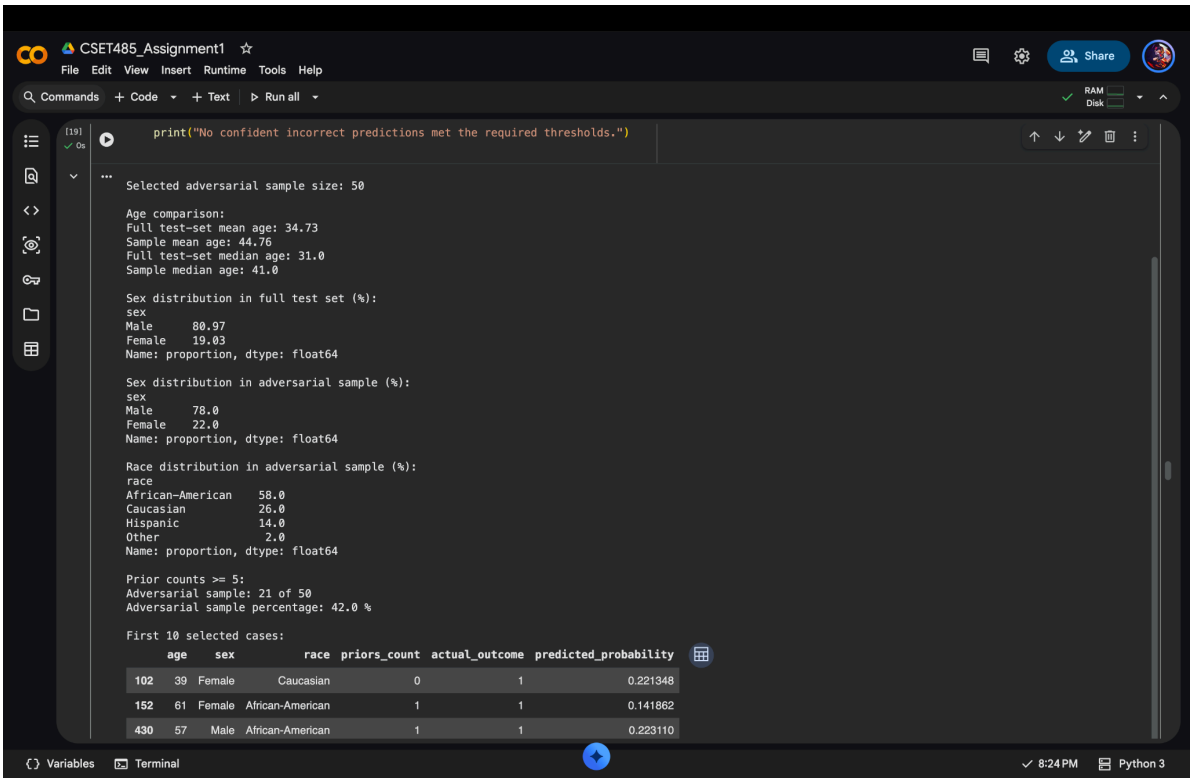
The table compares regression coefficients from the full model and the reduced model after deliberately omitting `priors_count`. The largest reported percentage change was 1196.7265% for `juv_misd_count`. Both models converged successfully. These changes demonstrate that coefficient estimates can be sensitive to the predictors included in a regression model; percentage changes should be interpreted carefully when the original coefficient is close to zero.

3.4 Adversarial Subset Analysis

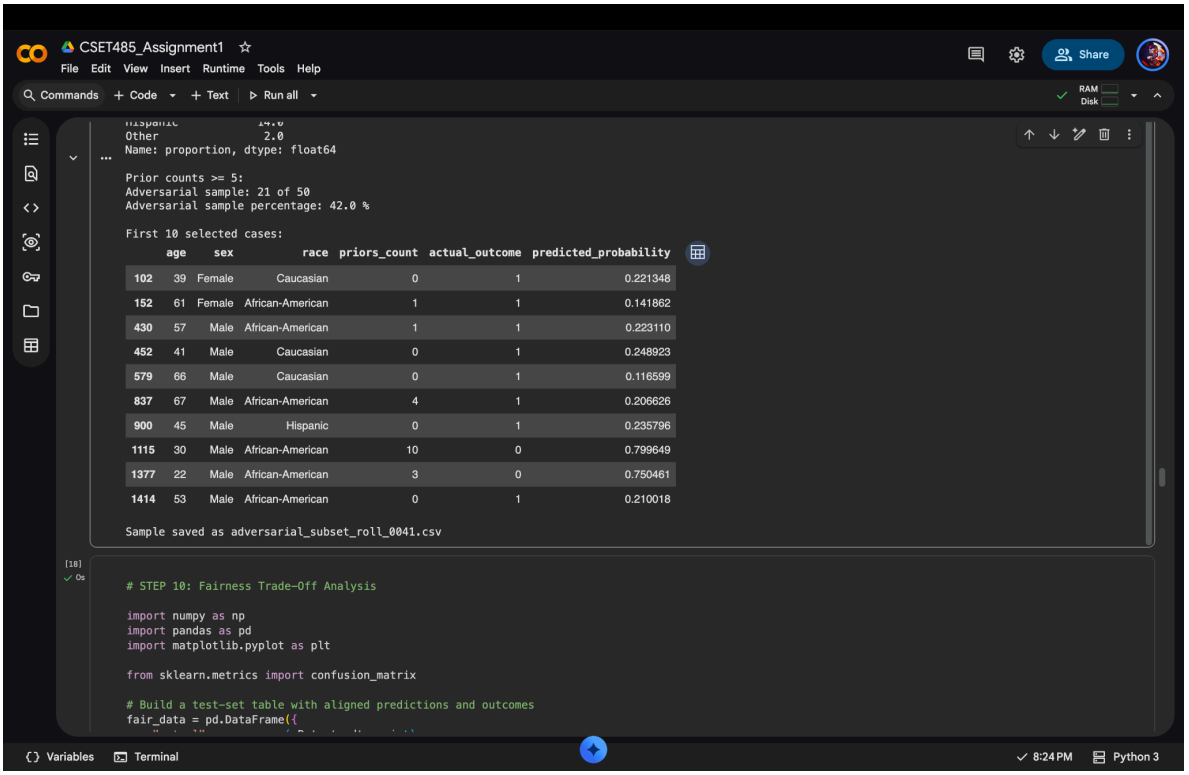
This part looked at the 50 selected cases. Out of these, 21 cases (42%) had five or more prior records, and 29 cases (58%) had fewer than five.

So a good number of these cases had multiple prior offences. Still, having a high prior count does not mean the person will definitely reoffend. Since the subset is small and was specially selected, I can't assume the same pattern holds for the whole population.

Screenshots :-



First screenshot — subset summary.



Second screenshot — first 10 selected cases.

Adversarial Subset Analysis Results

The analysis examined a selected subset of 50 cases. Of these, 21 cases (42%) had five or more prior records, while 29 cases (58%) had fewer than five. Because the subset was specially selected and relatively small, these findings should not be assumed to represent the entire dataset.

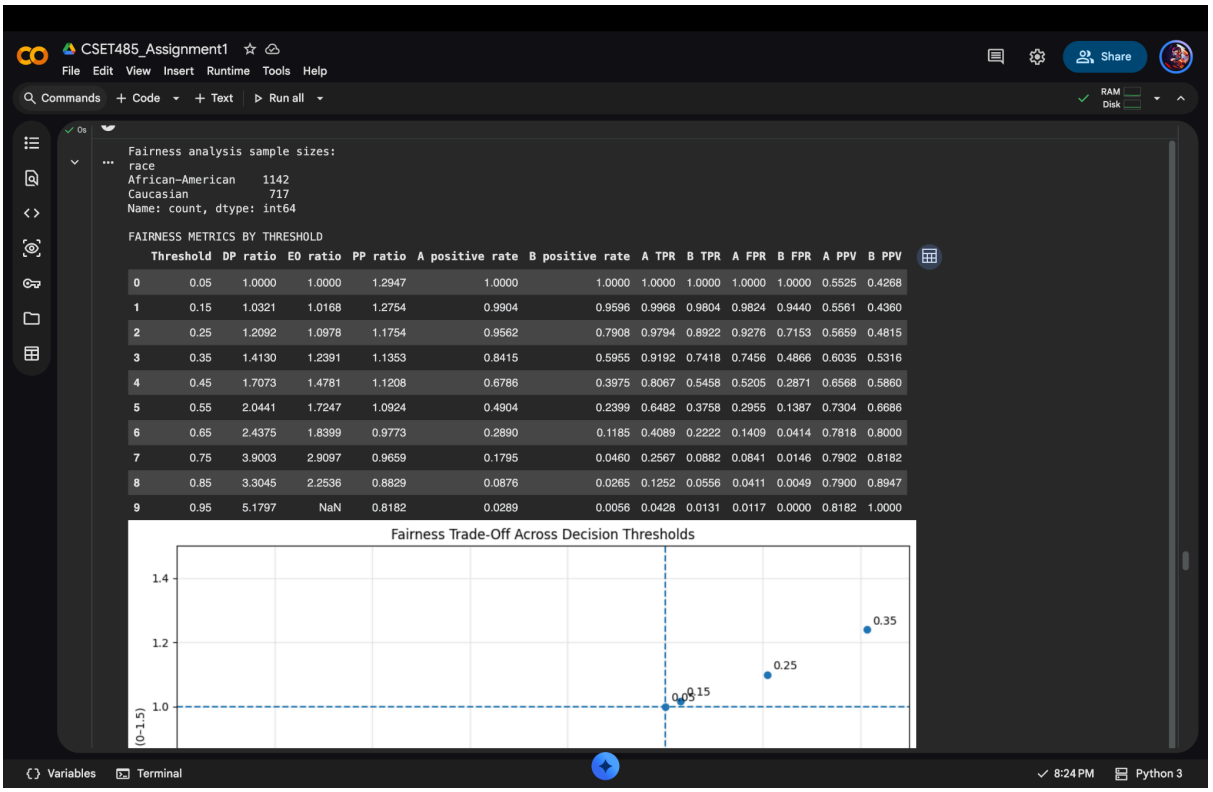
3.5 Fairness Evaluation

For fairness, I looked at demographic parity (DP), equalized odds (EO), and predictive parity (PP) at different thresholds.

At a threshold of 0.05, the DP ratio was 1.0000, EO ratio was 1.0000, and PP ratio was 1.2947. At 0.15, the values were 1.0321, 1.0168, and 1.2754. At 0.25, they were 1.2092, 1.0978, and 1.1754.

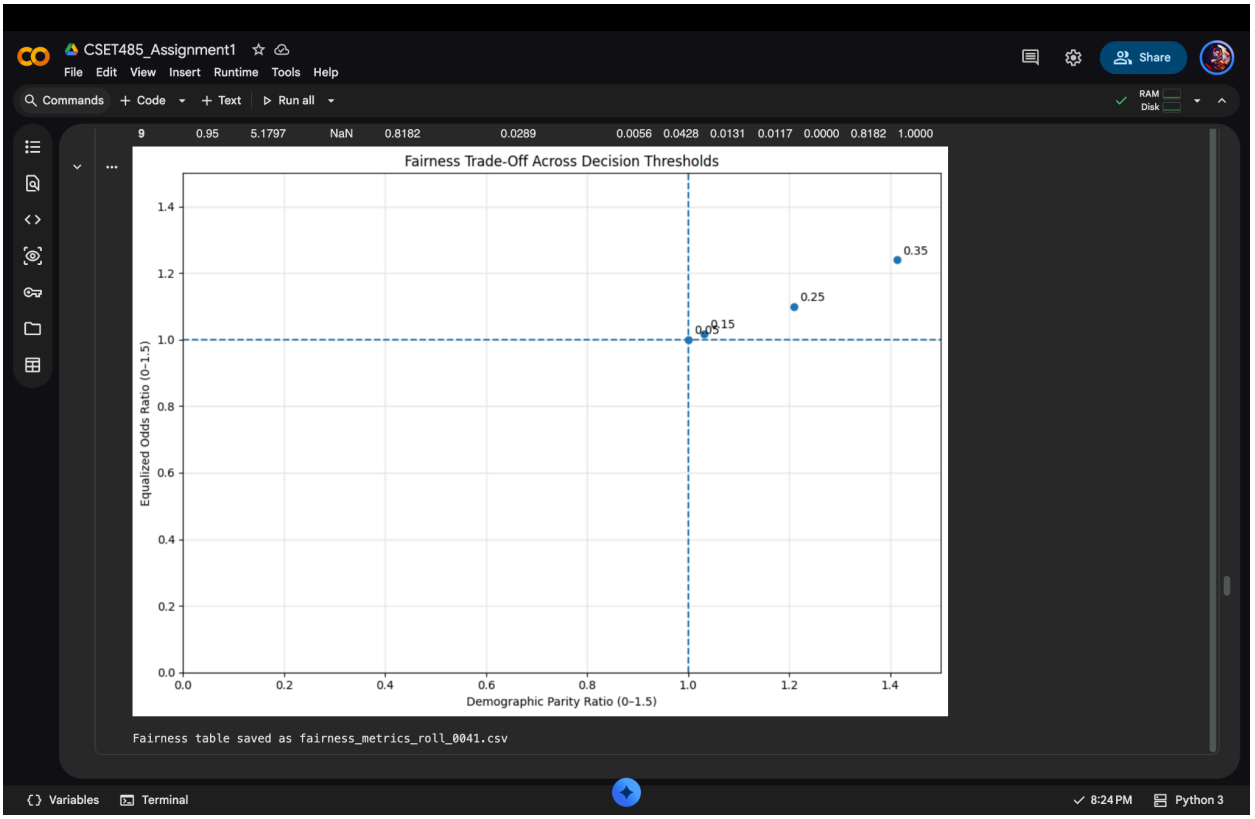
These results show that the fairness measures change depending on the threshold we choose. So no single ratio should be used alone to decide if a model is fair. We also need to consider the model's accuracy, error rates for each subgroup, uncertainty, and what happens when the model predicts wrongly.

Screenshot :-



Screenshot 1 — Fairness metrics table

It shows thresholds from 0.05 to 0.95 and the DP, EO, and PP ratios, along with group-level rates. This is the main evidence for your analysis.The table presents demographic parity, equalized odds, and predictive parity ratios at different decision thresholds, along with additional group-level performance rates.



Screenshot 2 — Fairness trade-off chart

It shows demographic parity ratios against equalized odds ratios at different thresholds. It supports your discussion of fairness trade-offs.The scatter plot illustrates the relationship between demographic parity and equalized odds ratios at selected thresholds. The dashed reference lines indicate a ratio of 1.0.

Part 3 — Descriptive Analysis Questions

COMPAS Recidivism Dataset | Roll Number 0041 | Answers grounded in the notebook outputs and reported tables

Question 1. Threshold-based approach versus direct logistic regression

The comparison between Linear Regression and Logistic Regression showed that the two models did not always classify the same people as high risk. In my test set of 2,165 cases, they agreed on 1,972 cases, which is an agreement rate of 91.09%. They disagreed on 193 cases, which is 8.91% of the test set. Out of these mismatches, 57 people were classified as low risk by Model A but high risk by Model B, and 136 were classified as high risk by Model A but low risk by Model B.

The main reason for this difference is that the two models use different prediction scales and decision rules. Linear Regression gives numerical predictions, and its classification depends on the median predicted value. Logistic Regression gives probabilities between zero and one, and in this assignment a person is classified as high risk when the predicted probability is at least 0.5. The median predicted value from my Linear Regression model was 4.5783, whereas Logistic Regression used a probability threshold of 0.50. These thresholds are on different scales: 4.5783 is a continuous regression prediction, while 0.50 represents a predicted probability of 50%. Therefore, the two models can classify some individuals differently.

This mismatch matters because a person who is high risk according to one model may be low risk according to the other. In criminal justice, this kind of disagreement could affect decisions about supervision, release conditions, or access to services. The 8.91% mismatch rate shows that the choice of model and threshold makes a real difference in practice.

Because of this, I would look at all 193 mismatched cases one by one, compare them with the actual outcomes, and check the false-positive and false-negative rates before recommending either model. Agreement between the models alone does not prove that they are accurate, reliable, or fair.

Question 2. Subgroup regression and differences in effects

My subgroup analysis used logistic regression on the full COMPAS dataset and then separately on African-American and Caucasian individuals. The full dataset had 7,214 observations, which included 3,696 African-American individuals and 2,454 Caucasian individuals. I focused on the coefficient for prior criminal-record count because it shows the association between prior records and the log-odds of recidivism.

In the full model, the prior-count coefficient was 0.1689, with a 95% confidence interval of [0.1551, 0.1826]. For the African-American subgroup, the coefficient was 0.1561, with a confidence interval of [0.1387, 0.1736]. For the Caucasian subgroup, it was 0.1871, with a

confidence interval of [0.1586, 0.2156]. All three coefficients were positive, and none of the confidence intervals included zero.

These results show that prior-record count was positively associated with the log-odds of recidivism in all three models. The coefficient was numerically larger for the Caucasian subgroup than for the African-American subgroup, but this does not prove that the difference between the groups is statistically significant. The confidence intervals overlap, so I would need a formal interaction test, or a direct test of the difference between the coefficients, to say whether the subgroup effects really differ.

The policy implication is that one overall coefficient can hide differences in how strong the association is across groups. If subgroup analysis is ignored, decision-makers might assume the same relationship applies equally to everyone. However, this does not mean that race causes recidivism. The models only describe associations in the observed data, not causal effects.

Question 3. Omitted-variable bias and coefficient changes

In my omitted-variable experiment, I looked at how the regression coefficients changed when one predictor was removed from the model. The variable I deliberately left out was `priors_count`, which is the number of prior criminal records. The purpose was to see whether the estimated relationships between the remaining predictors and recidivism were sensitive to the model specification.

The four predictors I examined were age, juvenile felony count (`juv_fel_count`), juvenile misdemeanor count (`juv_misd_count`), and juvenile other-offence count (`juv_other_count`). The largest percentage change was 1,196.7265%, for `juv_misd_count`. This suggests that its estimated coefficient was very sensitive to the removal of `priors_count`.

The percentage change is calculated as:

$$\text{Percentage change} = \frac{|\beta_{\text{reduced}} - \beta_{\text{full}}|}{|\beta_{\text{full}}|} \times 100$$

```
OMITTED-VARIABLE BIAS RESULTS
Variable deliberately omitted: priors_count

Coefficient comparison:
```

	Predictor	Full model coefficient	Reduced model coefficient	Percentage change
0	age	-0.5683	-0.4280	24.6965
1	juv_fel_count	0.0845	0.2311	173.3461
2	juv_misd_count	0.0144	0.1868	1196.7265
3	juv_other_count	0.0794	0.1483	86.8061

```
Full model converged: True
Reduced model converged: True
```

A large percentage change does not automatically mean a coefficient is unreliable. If the original coefficient is close to zero, even a small absolute change can look like a huge percentage. So I would also compare the signs of the coefficients, the confidence intervals, the statistical significance, and the absolute changes.

This experiment shows why a model should not be trusted without a sensitivity analysis. If an important predictor is left out, the remaining coefficients may partly pick up the effect of the missing variable. In a real application, this could lead to wrong interpretations and poor policy decisions. So I would compare several reasonable model specifications and note down which conclusions stay stable.

Question 4. Detecting a hidden confound in a real-world model

As a practitioner, I would use an independent external validation study along with an audit by domain experts to check whether an important confounding variable is missing from my recidivism model. A hidden confound is a factor that affects both a predictor and the outcome, which can create a misleading association. Since the variable is hidden or unmeasured, we can't always find it just by looking at the fitted model.

For example, access to stable housing and employment support may be linked to both a person's situation at the time of assessment and their chance of returning to the criminal justice system. If these factors are not measured properly, the model may give their effect to prior criminal records or demographic characteristics.

First, I would ask criminal justice researchers, social workers, and other domain experts to review the variables used in the model. They could point out factors that might be missing and suggest how to measure them. Then I would test the model on a separate dataset from another location or time period, using the same outcome definition and similar evaluation methods.

During this validation, I would compare calibration, false-positive rates, false-negative rates, and regression coefficients across groups and locations. I would also look at residual patterns to see if the errors change systematically with age, prior records, demographic groups, or other available characteristics. If adding a possible missing variable changed a coefficient by a lot, that would show the original model was sensitive to its specification.

My own analysis shows why this matters. After omitting `priors_count`, the percentage change for `juv_misd_count` was 1,196.7265%. A change this large means I should check the original coefficient and its absolute change before drawing conclusions.

This result demonstrates sensitivity to model specification, but it does not by itself prove that a hidden confounder exists. To investigate that possibility, I would examine additional relevant variables, compare alternative model specifications, and validate the findings on independent data.

No single test can prove that all hidden confounding has been removed. I would write down my assumptions, repeat the analysis with other specifications, and ask for independent review. In a high-stakes setting, predictions should support careful assessment and should not decide a person's freedom on their own.

Question 5. Characteristics of the 50-row adversarial subset

My adversarial subset analysis looked at 50 selected cases from the COMPAS data. The purpose was to see whether these cases shared characteristics that could help explain the model's errors. In my report, 21 of the 50 cases (42%) had five or more prior criminal records, and the other 29 cases (58%) had fewer than five.

This shows that multiple prior records were common in the selected cases, but not common enough to say that all 50 individuals had five or more. More importantly, prior-record count alone does not explain why a model would make a confident wrong prediction.

The assignment defines an adversarial case as a false positive with predicted probability of at least 0.75, or a false negative with predicted probability of at most 0.25. My report documents the 50-row subset and its prior-record distribution, but it does not confirm that all 50 rows meet these conditions. So I need to verify the selection criteria in my notebook before calling the subset 50 high-confidence errors.

Statistically, confident errors can happen when the model learns associations that don't generalize well to some cases. Logistic regression assumes a particular relationship between the predictors and the log-odds of the outcome. If important interactions, nonlinear relationships, or contextual factors are missing, the predicted probabilities may be poorly calibrated for some individuals.

To investigate this, I would compare the age distribution, gender breakdown, prior-record counts, actual outcomes, and predicted probabilities of the subset with those of the full test set. I would also calculate what proportion of the subset meets each adversarial condition and look for repeated patterns.The notebook identified 114 confident incorrect predictions among 2,165 test cases, representing 5.27% of the test set. This consisted of 57 confident false positives and 57 confident false negatives. The selected 50-row subset had a mean age of 44.76 years and a median age of 41 years, compared with a mean age of 34.73 years and a median age of 31 years in the full test set. These descriptive differences suggest that the selected subset was older than the overall test population. However, the subset was deliberately selected, so these differences should not be interpreted as evidence that age caused the model errors or that the subset represents the entire test set.

The main limitation is that a deliberately selected sample of 50 cases may not represent the wider population. It can help show weaknesses in the model, but it should not be generalized without further validation.

Question 6. Fairness trade-offs and their consequences for individuals

My fairness analysis compared demographic parity (DP), equalized odds (EO), and predictive parity (PP) across different decision thresholds. These measures capture different parts of fairness, so improving one does not necessarily improve the others.

At threshold 0.05, the DP ratio was 1.0000, the EO ratio was 1.0000, and the PP ratio was 1.2947. At threshold 0.15, the values were 1.0321, 1.0168, and 1.2754. At threshold 0.25, they were 1.2092, 1.0978, and 1.1754. These results show that changing the threshold changes the measured differences between demographic groups.

For a concrete example, consider two cases from my selected adversarial subset. Case 102 is a 39-year-old Caucasian female with zero prior records, an actual outcome of 1, and a predicted probability of 0.221348. At a decision threshold of 0.15, she would be classified as high risk, whereas at 0.25, she would be classified as low risk. Because her actual outcome is 1, the latter classification would be a false negative. Case 1115 is a 30-year-old African-American male with 10 prior records, an actual outcome of 0, and a predicted probability of 0.799649. He would be classified as high risk at both thresholds, resulting in a false positive. These examples illustrate how threshold choices can change individual classifications and how errors can affect people from different demographic groups.

The fairness ratios alone don't tell us which individuals would be affected, so I would use the actual predictions and outcomes from my test set to build this example properly. I would pick two real cases from different demographic groups and compare their outcomes under at least two thresholds.

My results also demonstrate a numerical fairness trade-off. At threshold 0.15, the demographic parity ratio was 1.0321 and the equalized odds ratio was 1.0168, both close to the ideal value of 1.00. However, the predictive parity ratio was 1.2754. At threshold 0.25, the predictive parity ratio improved to 1.1754, while the demographic parity and equalized odds ratios increased to 1.2092 and 1.0978, respectively. Thus, improving predictive parity at threshold 0.25 came with less favorable demographic parity and equalized odds ratios. These results show why a policymaker should examine multiple fairness metrics rather than select a threshold based on only one measure.

I would not choose a threshold just because its DP ratio is closest to 1.0. Instead, I would check false positives and false negatives for each group, think about the consequences for the people affected, and involve the relevant stakeholders. A responsible decision should record which fairness goal is prioritized and who might be harmed by that choice.

Question 7. Defending a decision threshold

A decision threshold is the probability above which the model classifies a person as high risk. Choosing it means balancing different fairness goals and checking how the resulting classifications affect people.

In my analysis, at threshold 0.05 the fairness ratios were DP = 1.0000, EO = 1.0000, and PP = 1.2947. At 0.15, DP was 1.0321, EO was 1.0168, and PP was 1.2754. At 0.25, DP was 1.2092, EO was 1.0978, and PP was 1.1754. Since a ratio of 1.0 means equality under the definitions used in this assignment, threshold 0.05 gave the DP and EO ratios closest to the ideal among these three examples. However, its PP ratio showed a bigger gap between groups.

I would consider threshold 0.05 as a candidate for further investigation if the priority were equal classification rates and the equalized-odds measure. But I would not automatically recommend it for deployment. A low threshold may classify many more people as high risk, which could increase unnecessary restrictions or interventions. So the threshold must also be checked along with accuracy, calibration, false-positive rates, false-negative rates, and group-specific results.

The group-specific rates also show why threshold selection requires careful evaluation. At threshold 0.25, the positive classification rate was 0.9562 for group A and 0.7908 for group B. The corresponding true-positive rates were 0.9794 and 0.8922. These differences show that the fairness ratios should be considered alongside the actual rates experienced by each group. Before choosing a threshold, I would also compare the false-positive rates, false-negative rates, and consequences of incorrect decisions for both groups.

At threshold 0.05, the positive classification rate was 1.0000 for both groups in my notebook. This means that all cases in both groups were classified as positive under the threshold rule. Although the DP and EO ratios were both 1.0000, this result illustrates why parity ratios must be interpreted alongside classification rates, error rates, and the practical usefulness of the model.

A critic could fairly argue that choosing a threshold just to bring two fairness ratios close to 1.0 does not guarantee that the model is useful or that the decisions are fair to individuals. Predictive parity is also far from its ideal value at 0.05. Also, the three thresholds I reported are only a part of the full sweep from 0.05 to 0.95.

My final recommendation would be conditional. I would compare all thresholds using the complete fairness table, choose a reasonable operating point based on the real consequences of errors, and validate that choice on separate data. I would also write down the strongest counterargument and explain why my chosen priority is still appropriate.

Question 8. Demographic parity versus equalized odds

If a policymaker had to choose between demographic parity and equalized odds for my COMPAS recidivism dataset, I would recommend prioritizing equalized odds, while still monitoring demographic parity and predictive parity. My reason is the consequence of wrong predictions in criminal justice, where an incorrect high-risk classification can affect a person's freedom and opportunities.

Demographic parity compares the proportion of people classified as high risk across demographic groups. Equalized odds focuses on differences in the true-positive and false-positive rates between groups. In this setting, I would give special importance to whether people who would not reoffend are wrongly classified as high risk, because such false positives can lead to unnecessary restrictions.

My results show how hard it is to satisfy all fairness goals at the same time. At threshold 0.05, the DP and EO ratios were both 1.0000, while the PP ratio was 1.2947. At threshold 0.15, the ratios were 1.0321, 1.0168, and 1.2754. At threshold 0.25, they were 1.2092, 1.0978, and 1.1754. These numbers show that the fairness measures change as the threshold changes, but they do not prove that any one threshold is the best.

A closer comparison of thresholds 0.15 and 0.25 illustrates this trade-off. At 0.15, the demographic parity ratio was 1.0321 and the equalized odds ratio was 1.0168, while predictive parity was 1.2754. At 0.25, predictive parity improved to 1.1754, but demographic parity and equalized odds moved farther from 1.0, reaching 1.2092 and 1.0978, respectively. I would prioritize equalized odds because differences in false-positive and false-negative rates can expose demographic groups to unequal error burdens. I would accept some difference in positive classification rates only after checking the actual group-specific error rates and the consequences for affected individuals.

Prioritizing equalized odds could mean accepting some difference between groups in the overall proportion classified as high risk. The reason is that equal error rates are more directly connected to stopping one group from facing unfairly high false-positive or false-negative rates. However, demographic parity is still important because large differences in classification rates can raise concerns about unequal treatment.

For example, at threshold 0.25, the demographic parity ratio was 1.2092, compared with 1.0321 at threshold 0.15. This indicates a greater disparity in positive classification rates under the ratio definition used in my analysis. Therefore, prioritizing equalized odds may require accepting a larger demographic-parity disparity, which should be evaluated alongside the actual false-positive and false-negative rates for each group.

I would not say this choice is correct for every situation. Policymakers should look at the actual group-specific error rates, predictive values, uncertainty, and real consequences of decisions before deployment. I would also do external validation and consult the communities affected. Most importantly, no fairness metric should replace individual assessment, transparent procedures, and meaningful human oversight in decisions about a person’s freedom.

5. Reflection on Model Evaluation, Fairness, and Deployment

This assignment helped me understand that checking a model's accuracy is not enough to say it is a good model. I used the COMPAS recidivism dataset to compare model classifications, look at regression results for different demographic groups, test omitted-variable bias, study a selected set of cases, and check fairness measures at different thresholds.

When I compared Linear Regression and Logistic Regression, I saw that two models can give different classifications for the same people. They agreed on 1,972 out of 2,165 test cases, but 193 cases were classified differently. This taught me that the choice of model, the prediction scale, and the threshold can all change the results.

In the subgroup regression, prior record count had a positive association with the log-odds of recidivism, but this does not mean it causes recidivism. I also learned that coefficients can change when other important variables are removed, so choosing the right variables is a big part of statistical analysis.

The fairness part felt the most important to me because predictions in criminal justice can affect a person's freedom. Demographic parity, equalized odds, and predictive parity each measure fairness in a different way. A threshold that looks good for one measure may not look good for another, so we can't decide fairness from a single ratio.

If I had to deploy a model like this, I would first test it on proper test data, check error rates for each subgroup, and make sure the data quality is good. I would also look at what happens with false positives and false negatives, write down the limitations of the model, and make sure the people using it understand how uncertain its predictions are.

I would not recommend using a recidivism prediction model as the only basis for decisions about someone's freedom. Human oversight, transparency, accountability, and regular monitoring are necessary. The model should also be checked in the setting where it will actually be used, keeping legal requirements and the harm from wrong predictions in mind.

Overall, this assignment taught me to read statistical results carefully, question assumptions, and think about the real-world effects of algorithmic decisions. A model should not be trusted as reliable or fair just because its numbers look good. It needs to be evaluated continuously.